

Assignment 5: Huffman Coding DESIGN.pdf:

Description of Program:

This program has an encoder as well as a decoder. The Huffman encoder will read in an input file, find the Huffman encoding of its contents, and use the encoding to compress the file. The Huffman decoder will read in a compressed input file and decompress it, expanding it back to its original, uncompressed size.

Files to be included in directory “asgn4” + pseudocode/structure:

- encode.c: This file will contain your implementation of the Huffman encoder.
- decode.c: This file will contain your implementation of the Huffman decoder.
- defines.h: This file will contain the macro definitions used throughout the assignment.
- header.h: This will contain the struct definition for a file header.
- node.h: This file will contain the node ADT interface. This file will be provided.
- node.c: This file will contain your implementation of the node ADT.
- pq.h: This file will contain the priority queue ADT interface. This file will be provided.
- pq.c: This file will contain your implementation of the priority queue ADT. You must define your priority queue struct in this file.

- code.h: This file will contain the code ADT interface. This file will be provided.
- code.c: This file will contain your implementation of the code ADT.
- io.h: This file will contain the I/O module interface. This file will be provided.
- io.c: This file will contain your implementation of the I/O module.
- stack.h: This file will contain the stack ADT interface. This file will be provided.
- stack.c: This file will contain your implementation of the stack ADT. You must define your stack struct in this file.
- huffman.h: This file will contain the Huffman coding module interface
- huffman.c: This file will contain your implementation of the Huffman coding module interface.
- Makefile
 - • CC = clang must be specified.
 - • CFLAGS = -Wall -Wextra -Werror -Wpedantic must be included.
 - • make should build the encoder and the decoder, as should make all.
 - • make encode should build just the encoder.
 - • make decode should build just the decoder.
 - • make clean must remove all files that are compiler generated.
 - • make format should format all your source code, including the header files.

- README.md
 - In proper Markdown syntax.
 - Describe how to use your program and Makefile
 - Lists and explains any command-line options that program accepts.
 - Reports any false positives by scan-build
 - Notes down any known bugs or errors in this file as well for the graders.
- DESIGN.pdf
 - This document must be a proper PDF.
 - Describes design and design process for program with enough detail such that a sufficiently knowledgeable programmer would be able to replicate implementation.